ME 2110 - Section A11

Final Report

Team 5

Wael Al Ghanami

Brandon Kluge

Sean Osterwise

Joshua Wu

Submitted to Mr. Nathaniel DeVol

TA: Oliver Stephan

Date: 11/16/2022

Abstract

In this report, the entire process of constructing the robot, from design to fabrication, is discussed. Multiple robot designs were drafted from the morphological chart and then using an evaluation matrix the best possible design was determined. The final design features a large motor powered base, with wings, a motor arm, and pneumatic pushes for the soldiers and arrows. It also places Legolas into Mount Doom using a solenoid triggered mousetrap mechanism. The construction focused on emphasizing time for testing the robot rather than spending every available moment building it. The process was a successful one, and the final design was successful due to the planning which was done at the start of the process. Making a robot was not a simple task, and required many new skills, from programming to laser cutting to be learned in order for everything to function properly.

I. Introduction

The goal of this project is to design a robot system that can successfully perform all 5 tasks in the *Lord of the Rings* themed robot competition, whilst simultaneously avoiding disqualification. The solutions for this project were limited by specification restraints including a runtime under 40 seconds, a budget under $100, power from a list of pre-specified sources, and fitting within a 12" x 24" x 18" space. Table 1 shows these along with all other specifications considered in the design. Due to these restraints, several design challenges were considered during the ideation of potential solutions. These challenges included reaching the top of Mt. Doom, determining when to deploy troops and arrows, pushing orcs, lifting Legolas, and lastly, prioritizing and distributing each of the limited power sources accordingly between all of the tasks. This report discusses a proposed solution to the problem which should successfully complete all the tasks with maximum accuracy, and in the shortest time possible.

II. Problem Understanding

Customer needs were compared against engineering specifications to identify the most important requirements for this project. The customer is defined as the design team.

The nature of this project dictates that the most crucial task is to not get disqualified, therefore functions directly related to disqualification were weighted a higher relative importance than those of achieving specific tasks. The House of Quality for this robot (Fig. 1) which evaluated the importance of the customer needs and engineering requirements suggests that the most vital engineering specifications are movement accuracy (24% relative importance) and sensor detection accuracy (13% relative importance). Both accurate movement and sensor detection contribute to successful positioning, and successful positioning prevents disqualification. The two specifications synergize effectively, as investing in better sensors improves movement accuracy. There are some trade-offs within these customer needs, though, as there is an inverse synergy with program time, price, and robot size, which all have strictly enforced ceilings. The challenge would be to optimize between accurate movement and price, size, and run time.

The force of components has a relative importance of 9%; it governs factors such as the safety of the track and reliability of movement. The sources of force are limited: 2 DC motors, pneumatic actuators, solenoids, mousetraps, rubber bands, and gravity. An analysis of the force and reliability of each source would aid in correctly allocating power sources to tasks. For instance, it would be wise to use DC motors to achieve tasks which require precise positioning, while stored elastic energy could be used for tasks with a large margin of safety, such as knocking orcs down. With the caveat of causing no damage to the track, a factor of safety of at least 1.2 should be considered to avoid fracture or deformation of the wooden track.

III. Conceptual Design

The main functions required of the system were moving the orcs (cubes) to specific zones, deploying Legolas and soldiers in the battle station, launching arrows in other battle stations, and placing a ring on a post (Fig. 2). Table 2.1-2.3 demonstrates the different designs that were considered for all of the different functions.

Both motors and mouse traps were considered to drive the system. Motors would turn gears turning axles/wheels. Mouse traps would pull a string, causing a short, immediate torque around an axle (Fig. 3). Of the two options considered, motors were more practical for this machine. The motor maxes at 60 rpm (1 rotation per second) and produces 0.5N-m of torque. A motor offered more precision in its movements, and also allowed the system to move backwards if needed. According to the calculation in Figure 4, it will take about 13 seconds for the system to move to its position leaving 27 seconds to complete all the tasks.

The plan for moving the orcs was to deploy a pair of wing-like attachments on each side of the system (Fig. 5). Originally, this system would have dropped a one piece wing, then pushed the orcs as the system approached Mt. Doom. The current design had the wings folded over at the top to provide more distance to push the orcs. The ring mechanism released these wings in addition to its other function. A pneumatic actuator that pushes the orcs at each position as the system moves past was also considered, but this technique would both require a valuable component in the actuators, and be difficult to time correctly mechanically so the wings were considered the better option.

The arrow and soldiers were stored in a tube made of PVC pipe, with the soldiers held up by a gate and the arrow held in by a solenoid. A rolling switch was used to identify the correct zone, then a pneumatic actuator opened the soldier gate (Fig. 6). After a delay, the arrow was dropped by the solenoid down the same tube. The rolling switch would then wait for the correct zone a second time to deploy the Legolas mechanism. A mousetrap powered "catapult" (Fig. 7) would then lift Legolas into the zone directly from his podium. Originally, this was planned to be a lift or elevator of some sort, but the mousetrap design was significantly simpler and more consistent. This entire process will take maximum of 20 seconds

To detect the zone, a rolling switch was used. 4 knobs were mounted at different fixed heights on the centerpiece. Each knob identified a zone. The rolling switch was placed at the height of the desired switch of the desired zone, and pressed just before the soldiers and arrows needed to be deployed. A color sensor was considered for this task, but a drawback with the color sensor is that it would have likely been affected by variations in the lights in the performance area, so ultimately it was decided against.

For ring delivery, the ring needed to be placed on a post on the peak of Mt. Doom. The ring was held by friction, fit between two surfaces at the tip of the mechanism. An arm attached to a motor swung the ring out towards Mt. Doom (Fig. 8). A motor was chosen for this task because the arm movement needs to be slow and precise. The ring could have slipped and become a projectile otherwise. Once the ring was upon Mt. Doom, the system retreated. The normal force of the pole is greater than the friction force holding the ring in place, so the ring remains on the pole. To retreat, the system leaves in the same way it entered, with the motor drive in the opposite direction. This drive system was operated by having the motor attached to a gear and axle. This is a quick and efficient way to move the system away from Mt. Doom. Figure 9 demonstrates what the robot was theorized to look like when all of the components are put together, and Figure 10 is a more accurate depiction based on final engineering decisions.

IV. Design Overview and Fabrication

The chassis modeled in Figure 11 total dimensions are 10”x16’’ to allow enough distance for the robot to move inside the playing field. The width is 10’’ to ensure that there is sufficient space for the orc pushers. The middle section of the chassis was cut to allow the robot to fit around Legolas. 0.75” plywood was screwed to the base to provide extra support to the MDF. 0.5’’ aluminum axles were chosen as they are strong and light. Two 20-tooth gears connect the motor to the back wheels that are mounted by 3D printed axle holders.

The ring is placed with a motorized arm (Fig. 12). A small motor was placed on a stand close to the height limit of the robot and is connected to the ring arm via gears. When the robot reaches Mt. Doom, the arm revolves 200 degrees, placing the ring. The most important specifications that need to be reached with this arm is the level of friction which holds the ring.

The orc pushers are 5mm thick square sheets of wood cut diagonally across the corners (Fig. 13). The ends were cut and hinged together to unfold, and blocks were glued on corners to ensure a bend no less than 90 degrees. Key specifications for this design are the maximum height above ground; anything higher than 2 inches will pass above the orcs.

The Soldier and Arrow mechanism (Fig. 14) are a tube with a ramp at the end that is pushed down by pneumatic actuators. The key specification for this mechanism is the speed it can deploy; 3-4 balls must be launched in a 2.5 second window. The tube-ramp system is made of PVC. A mounted actuator will lower the ramp, allowing the troops and arrows to freely roll into the desired zone. Legolas will be moved into Mt. Doom by a cutout piece which is connected to a mousetrap. The piece will have a cutout wider than the Legolas pole but less than the diameter of Legolas. A solenoid will hold the cutout piece down and when it is triggered it will spring up and deposit Legolas in Mt. Doom without needing to knock Legolas off the pole.

The robot follows a fairly linear order of operations shown in the Fig. 15 flowchart. The robot begins by slightly jolting the ring arm to displace the orc pushers from their folded position. It then drives forward to Mt. Doom, collects Legolas, and places the ring. The correct zone is detected by a long arm switch with adjustable height for each color. When the beacon hits the switch, a pneumatic actuator pushes forward. This drops a ramp to get the soldiers and triggers the large solenoid to release Legolas. Later, the small solenoid retracts, dropping the arrows into the next enemy zone. Upon release of the arrows, the robot backs away and stays static between the shire and Mt. Doom until time runs out.

V. Alternative Designs

The first alternative to the chosen design drawn in Fig. 16 uses moustraps instead of motors to drive the robot and move the arm with the ring. The ring itself is released with a solenoid trigger at the end of the arm. On contact with Mt. Doom, a second solenoid deploys arms which sweep the orcs to their zones. Legolas will be raised with a fork and motor mechanism like the one used in the chosen design, although the lift uses a motor instead of a counterweight. The robot will use switches which hit the tabs on Mt. Doom to know the orientation. This design scored a 0.7083 relative total in the design evaluation matrix (Tbl. 3), the main liabilities being poorer movement accuracy, force of components, and setup time which all scored two of four mainly caused by the mousetraps unreliability. The second alternative shown in Fig. 17 returns to motor-based driving. In this design, the ring extends over Mt. Doom using a rail which functions similar to an aerial ladder on a fire truck. Actuators push the orc-arms out, pushing the orcs while Legolas is catapulted into the soldier-delivery using an internal mousetrap. This design also uses the tabs on Mt. Doom, but these levers are purely mechanical using a bevel or miter gear. This design scored a 0.7292 relative total, the main issue being the unpredictability of the catapult or the actuators. The final alternative in Fig. 18 kicks the arm with the ring onto Mt. Doom using a single actuator. The ring release and motor drive are consistent with the chosen design and the Legolas lift is similar to the one used in alternative 1. The orc pushers will extend by actuator push as well. This design scored a 0.6458 relative total, requiring the most additional material for the weakest outcome.

VII. Performance Results

The final version of the robot was capable of scoring all elements of the challenge, resulting in a max score of 930 points. In sprint one the robot placed second out of 64 teams, and it placed sixth in sprint 2. In the final competition it placed 26th at the end of the second round, and was eliminated in the 3rd due to an alignment miscue which caused it to get stuck on the Legolas pole. While the final performance was mildly disappointing, ultimately the robot still managed to be fairly successful across the three competitions.

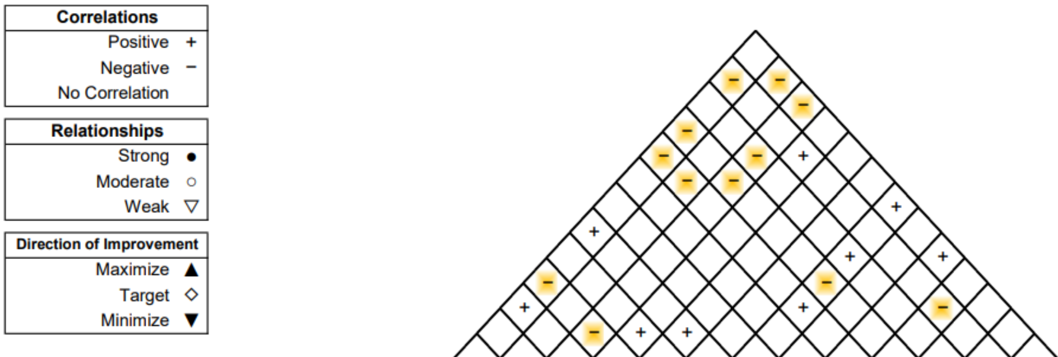
VIII. Conclusion

Building a robot to complete all aspects of the challenge was a difficult task. Starting out with no framework and ending up with a complete and functioning robot was a rewarding journey which provided many beneficial lessons. As Table 4, which is a Bill of Materials, shows, the robot was able to be built for a total of around $33, which is very economical for the entire process. Generally, the first design for mechanisms did not work perfectly, so figuring out ways to slightly modify mechanisms to work properly without re-designing the entire part was a skill which progressed rapidly due to the challenge. The final version of the robot turned out well, as it was capable of doing everything asked by the challenge, even if the final competition was not as successful as it could have been. Moving forward, all of the problem solving and fabrication skills developed during ME 2110 will be greatly beneficial to provide both a foundation for future Georgia Tech engineering courses and the work world upon graduation.

Appendix

Table 1 - Specification Sheet

Specification List for Robot				
Changes	D/W	Requirement	Responsibility	Source
		Geometry		
09/24/2022	D	Max Robot Size: 12'' x 24'' x 18''	Design Team	ME 2110 Specs
09/24/2022	W	Wheel Radius: 2''	Design Team	Team
09/24/2022	D	Hollow Interior: 10'' x 20'' x 14''	Design Team	HOQ
		Operation		
09/24/2022	D	Max Run Time: 40 seconds	Design Team	ME 2110 Specs
09/24/2022	D	Autonomous: Yes	Design Team	ME 2110 Specs
09/24/2022	W	Max Setup Time: 150 seconds	Design Team	Team
09/24/2022	W	Robot Sensors: color sensor, ultrasonic	Design Team	HOQ
		Power		
09/24/2022	D	AC Adapter: 1.5 A	Design Team	ME 2110 Specs
09/24/2022	D	AC Adapter: 3 A	Design Team	ME 2110 Specs
09/24/2022	W	Elastic Energy: Mousetraps & Rubber Bands	Design Team	ME 2110 Specs
		Material		
09/24/2022	D	Body: MDF	Design Team	Team
09/24/2022	D	Gears: MDF	Design Team	Team
09/24/2022	W	Axle: Steel	Design Team	Team
		Carry Capacity		
09/24/2022	D	Ring: 4'' OD	Design Team	ME 2110 Specs
09/24/2022	D	Troops: 1.57'' diameter spheres	Design Team	ME 2110 Specs
09/24/2022	W	Robot Mass: 7 lbs	Design Team	Team
		Cost		
09/24/2022	D	Bill of Materials: less than $100	Design Team	ME 2110 Specs



		Column #	1	2	3	4	5	6	7	8	9	10	11	12	13
		Direction of Improvement	▼	◇	▼	▲	◇	▼	▼	▼	▲	◇	▲	◇	▲
Category	Weight	Engineering Requirements / Customer Requirements (Explicit and Implicit)	Robot Size	Program Time	Price	Aesthetics Jury	Efficient Wiring	Set-up Time	Force of Components	area of movement	Movement Accuracy	Reachable Height	Durability	Storage Capacity	Sensor Detection Accuracy
Safety	10	Safety	▽				○		●		▽	▽	●		
Consistency	10	Operate Autonomously		●							●				●
	5	Ease of Operation	○				○	●							
	7	Build Quality			●	▽	●		○				●		
Complete tasks	10	Launch System									▽				
	8	End in Correct Position	▽	●							●				○
	5	Fire Arrows							○		●	○		●	●
	9	Deliver Ring							○		●	●		●	○
	7	Defend Against the Orcs	▽						○		●				○
	6	Collect Legolas							○		●	○		●	○
	7	Deploy Troops							○		●	○		●	●
	6	Deploy Legolas							○		●	○		●	●
Prevent Disqualification	10	Don't Enter Other Zones	▽							●	●				
	10	Be Within Budget	▽		●	○									
	10	Finish Within Time		●											
	10	Fit in Box	●		▽										
		Target	12x24x18 inches	<=40s	<=$100	100%	No interference with parts	3 mins	0.02lbf	Within safe zone	±0.5 inches	23.13 inches	5lbs	5x1.57inch diameter balls +	±5%
		Max Relationship	9	9	9	3	9	9	9	9	9	9	9	9	9
		Technical Importance Rating	150	252	163	37	108	45	231	90	632	163	153	297	342
		Relative Weight	6%	9%	6%	1%	4%	2%	9%	3%	24%	6%	6%	11%	13%
		Weight Chart													

Figure 1 - House of Quality

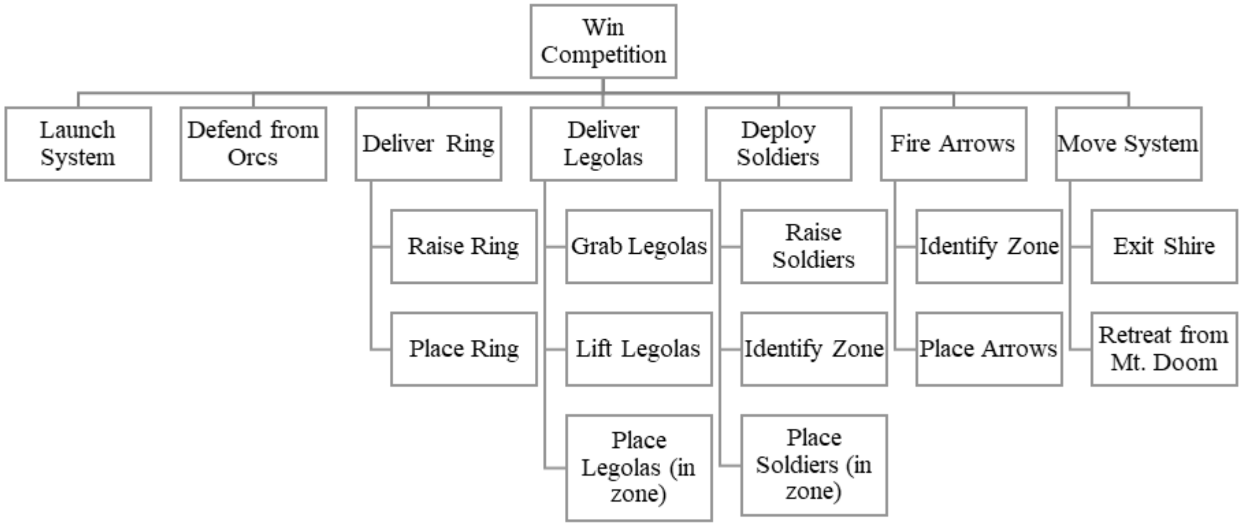


Figure 2 - Function Tree

Table 2.1 - Morphological Chart (Launch System, and Legolas)

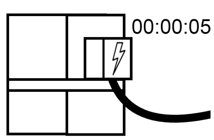
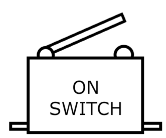
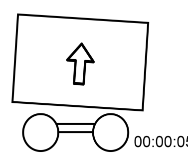
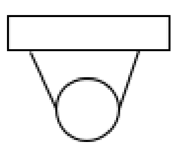
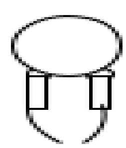
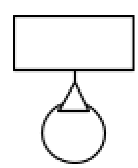
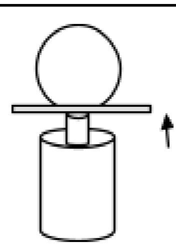
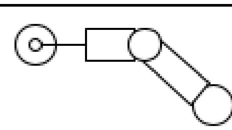
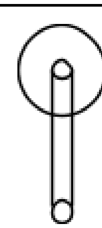
Function	Solution 1:	Solution 2:	Solution 3:
Launch System	 Timer countdown from when plugged in	 Button press	 Sensor and timer when to start when box is lifted
Grab Legolas	 Grab From Sides	 Lift with Fork	 Pick up from above
Lift Legolas	 Actuator	 Arm and System movement	 Lift with motor and spool

Table 2.2 - Morphological Chart (Orcs, Soldiers, and Ring)

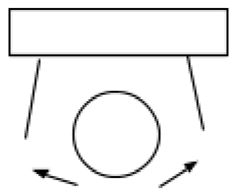
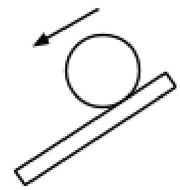
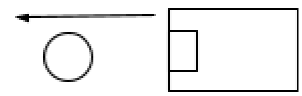
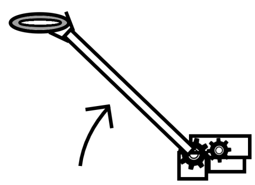
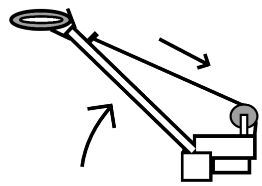
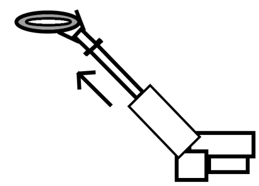
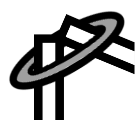
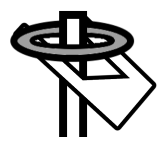
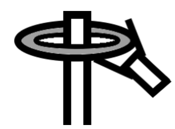
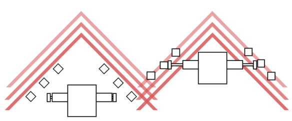
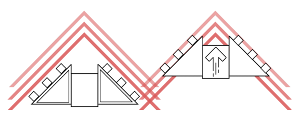
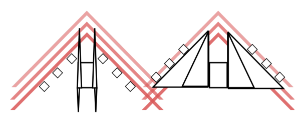
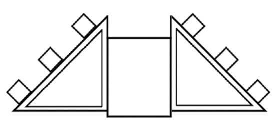
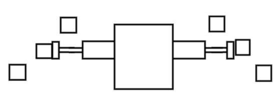
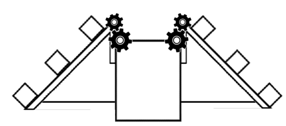
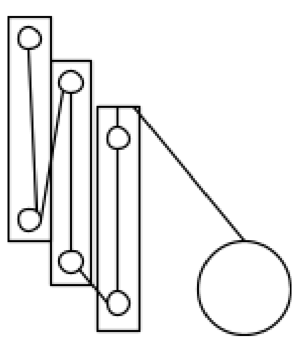
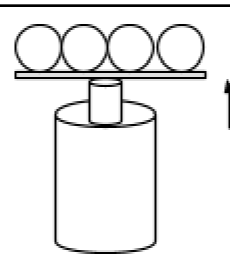
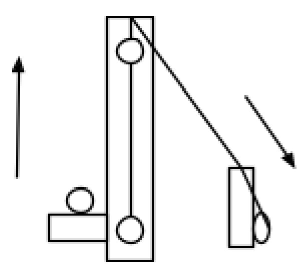
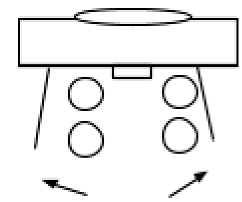
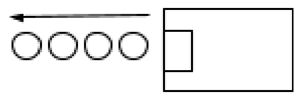
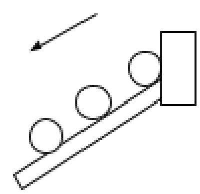
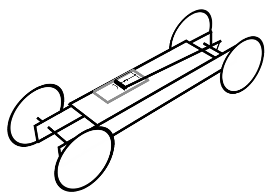
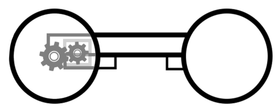
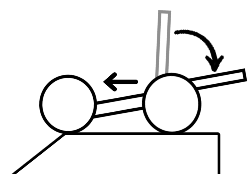
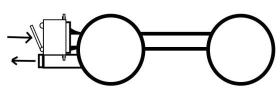
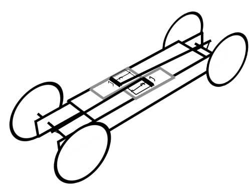
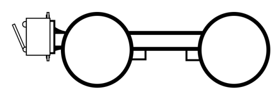
Function	Solution 1:	Solution 2:	Solution 3:
Place Legolas	 Release	 Roll off	 Launch horizontally
Raise Ring	 Motor attached to arm	 Spool of string	 Actuator
Place Ring	 Bar	 Fork	 Claw release
Push Orcs	 Actuator pumps at each orc location	 Sides of robot push orcs	 Sides of robot push orcs
Correct Zone	 Sides of robot shaped like wings	 Actuator pump distance limited	 Motor movement restricted
Raise Soldiers	 Vertical slide system	 Actuator	 Mousetrap

Table 2.3 - Morphological Chart (Drive System, and Soldiers)

Function	Solution 1:	Solution 2:	Solution 3:
Identify Zone	 Color sensor	 Timer	 Motion sensor
Place Soldiers Place Arrows	 Release from platform	 Shoot horizontally	 Slide down
Exit Shire	 Mousetrap Car	 Drive system from motors	 Shifting momentum
Retreat from Mt. Doom	 Actuator push w/ button/lever	 Mousetrap	 Drive system w/ button/lever

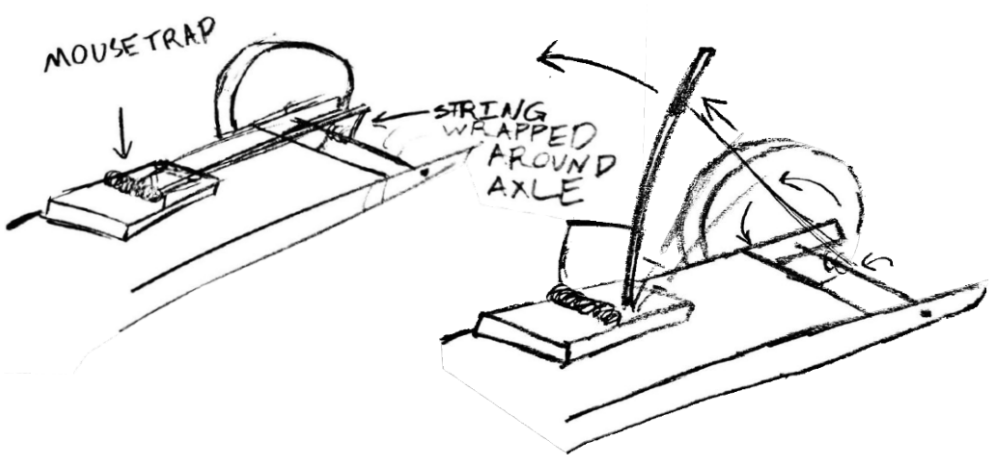


Figure 3 - Mousetrap Drive

Mouse trap $= .815\ J$

Robot has to move ≈ 21 in

robot estimated weight $- 6\ lbs = 2.72\ kg$

3 mouse traps $= .815\ J + .815\ J + .815\ J / 6 = .00245\ kJ$

$$KE = \frac{1}{2}mv^2 \rightarrow v = \sqrt{\frac{2KE}{m}}$$

$$\text{so } v = \sqrt{\frac{2(.00245\ kJ)}{2.72\ kg}} = .042\ m/s$$

$$\frac{.042\ in}{1\ s} \times \frac{39.37\ in}{1\ m} = 1.65\ in/s$$

$$21\ in \times \frac{1\ s}{1.65\ in} = \boxed{12.73 \text{ seconds to move robot}}$$

Ramp was not factored in, but it can be assumed the potential energy from the ramp will also help reduce time to move so it will be under 12.73 seconds, but for sake of estimate 12.73 seconds will be used

Figure 4 - Calculation for Time System Needs to Move

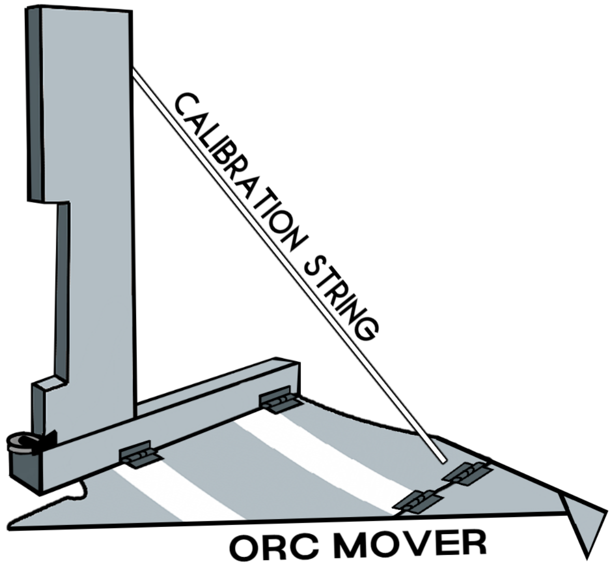


Figure 5 - Wing-like Attachments

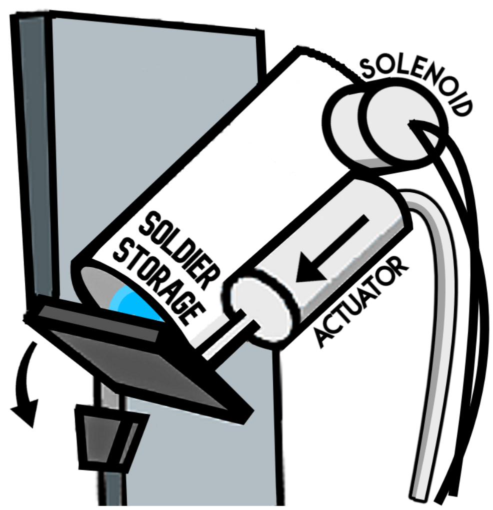


Figure 6 - Soldier Queue

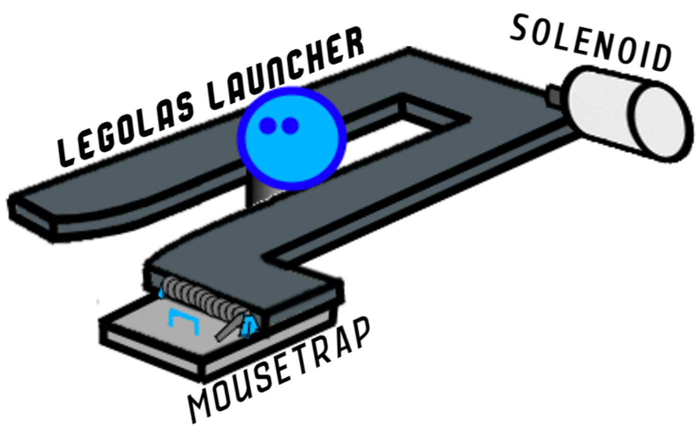


Figure 7 - Legolas Launcher

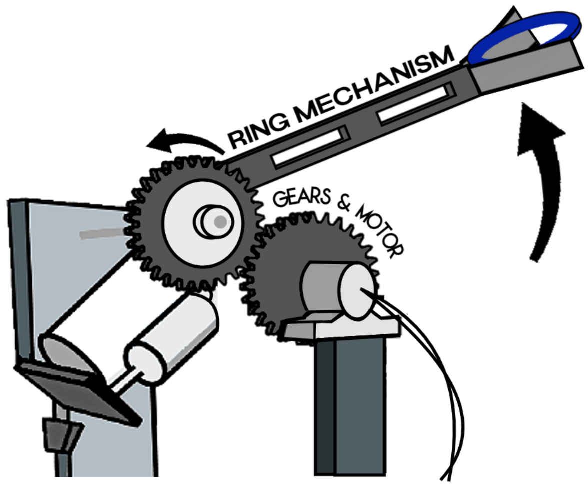


Figure 8 - Ring Delivery

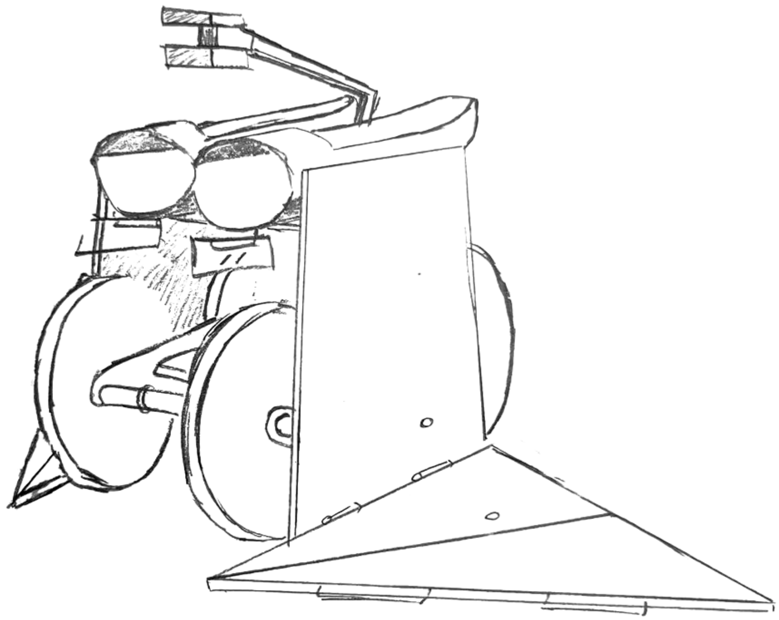


Figure 9 - Final Sketch of Robot Design (Theorized)

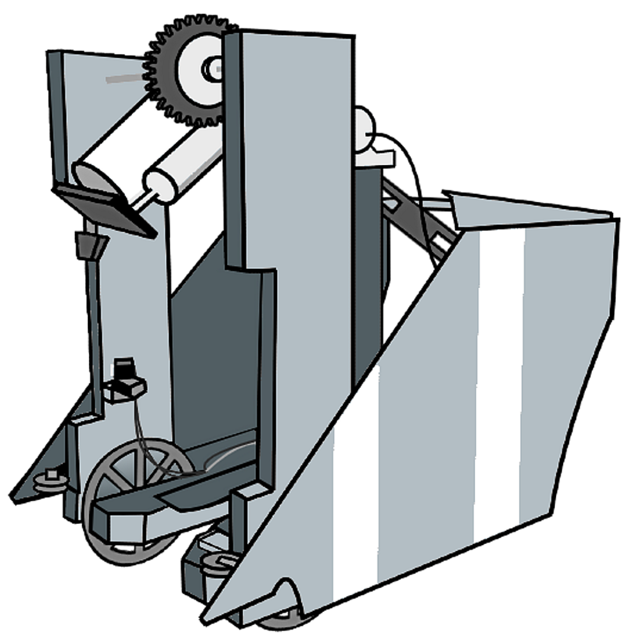


Figure 10 - Final Sketch of Robot Design (Accurate)

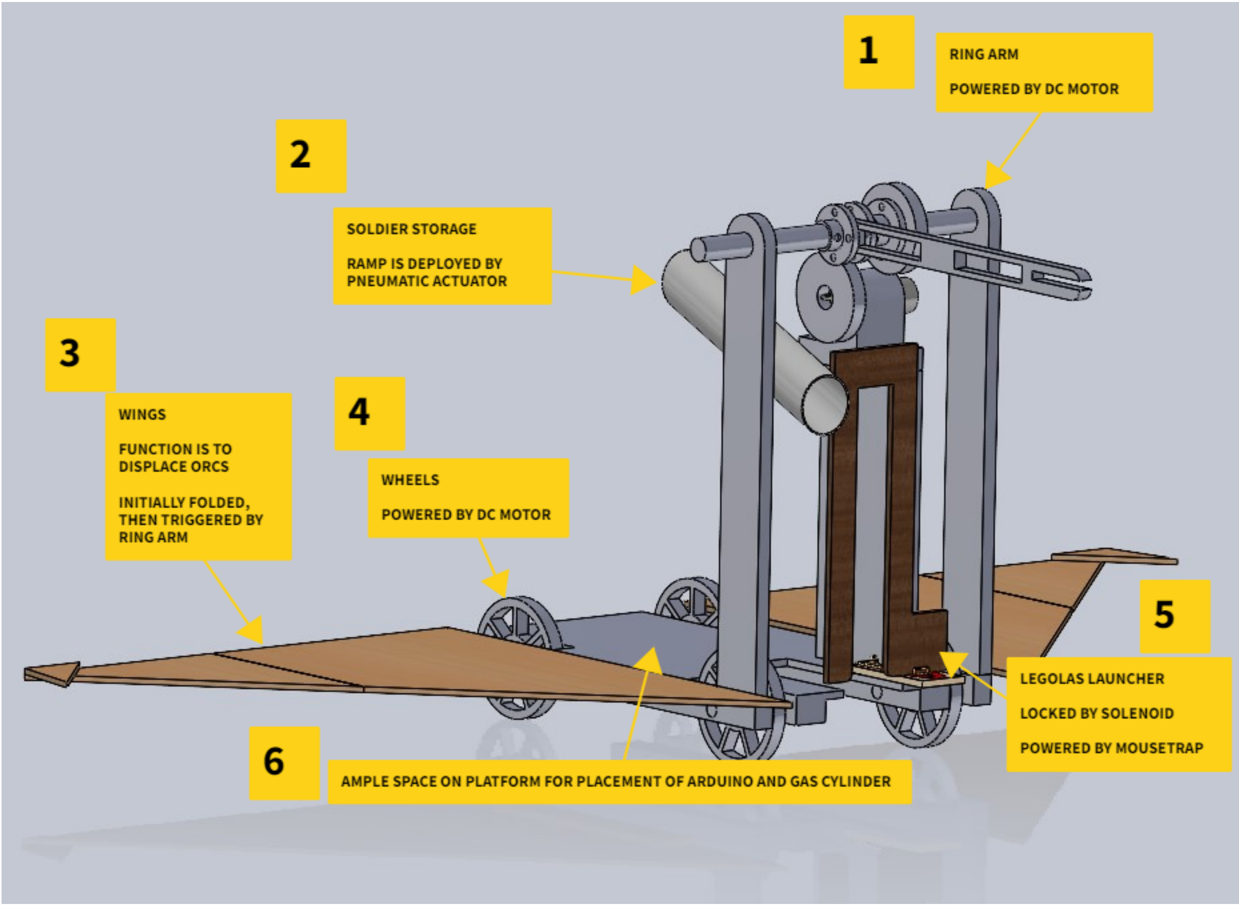


Figure 11 - Annotated CAD Assembly

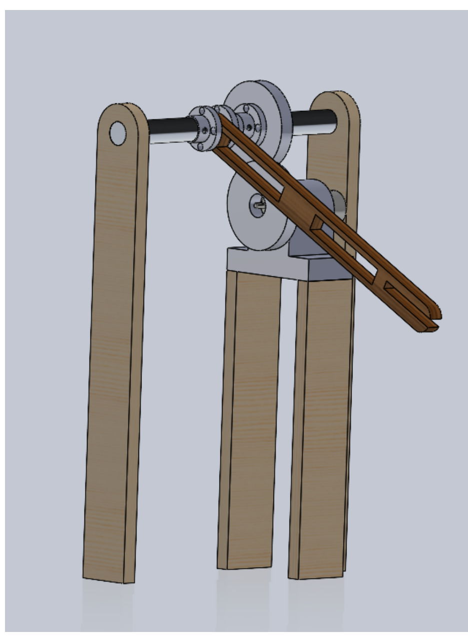


Figure 12 - CAD of Ring Placement Arm

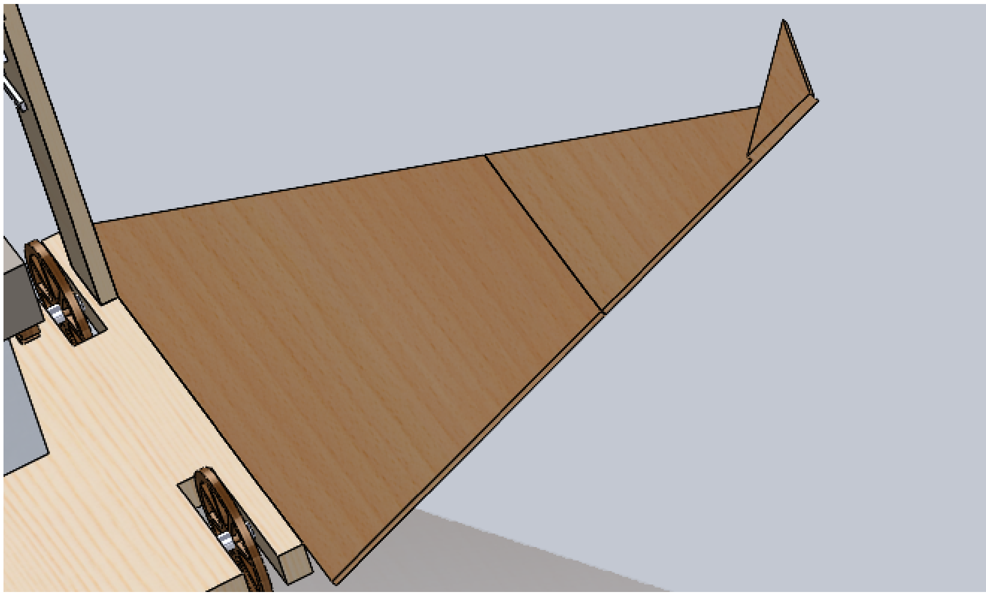


Figure 13 - CAD of Wing-like Orc Pushers

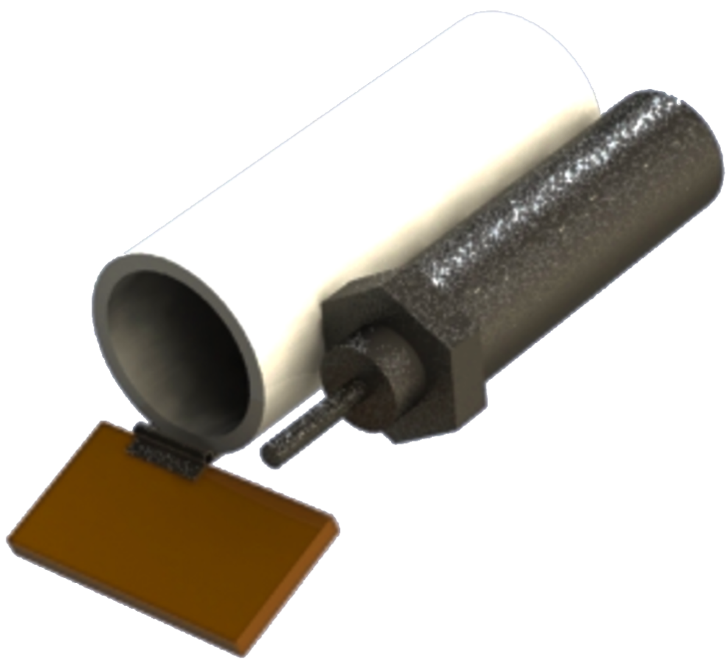


Figure 14 - CAD of Soldier Delivery

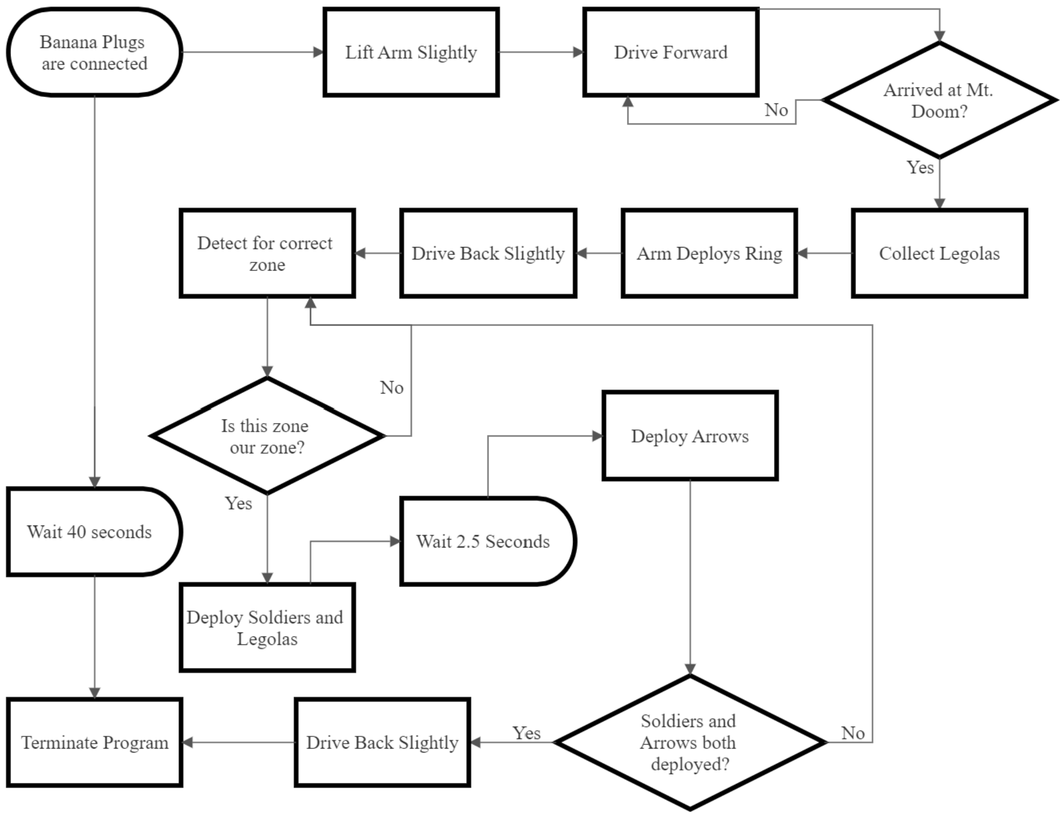


Figure 15 - Flowchart for Program

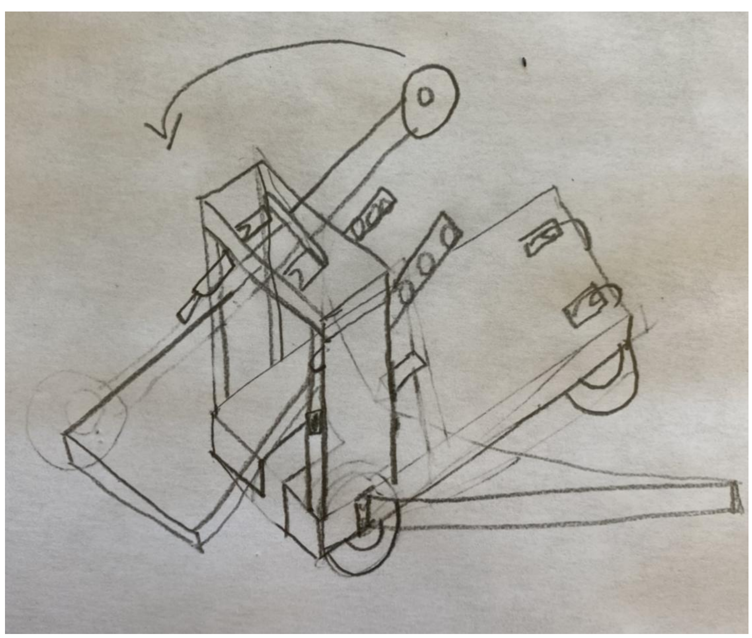
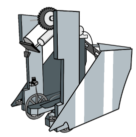
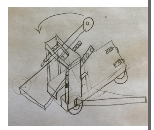
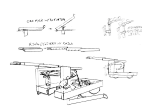
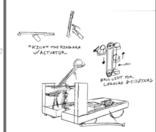


Figure 16 - Alternative Design #1, Mousetrap and Pneumatic Powered

Table 3 - Evaluation Matrix for Chosen and 3 Alternatives

Criteria	 **Chosen Design**	 **Alt. 1**	 **Alt. 2**	 **Alt. 3**
Robot Size	4	3	3	4
Efficient Wiring	3	3	3	2
Setup Time	4	2	3	2
Force of Components	3	2	2	2
Area of Movement	4	3	4	4
Movement Accuracy	4	2	2	2
Reachable Height	3	4	3	4
Durability	3	3	3	1
Storage Capacity	2	3	3	4
Sensor Accuracy	3	4	4	2
Appearance	4	2	3	2
Price	4	3	2	2
Total	41	34	35	31
Relative Total	0.8542	0.7083	0.7292	0.6458
Rank	1	3	2	4

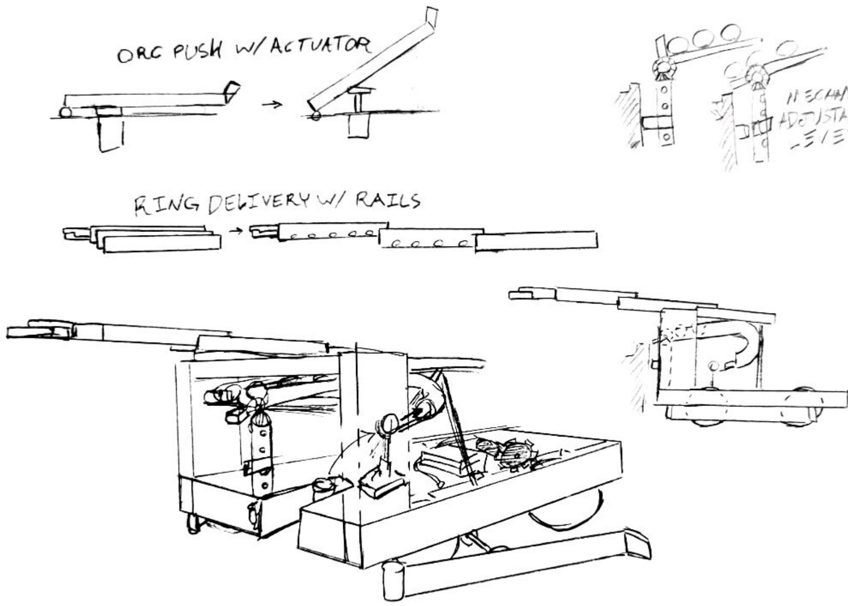


Figure 17 - Alternative Design #2, Rail-Ring Delivery

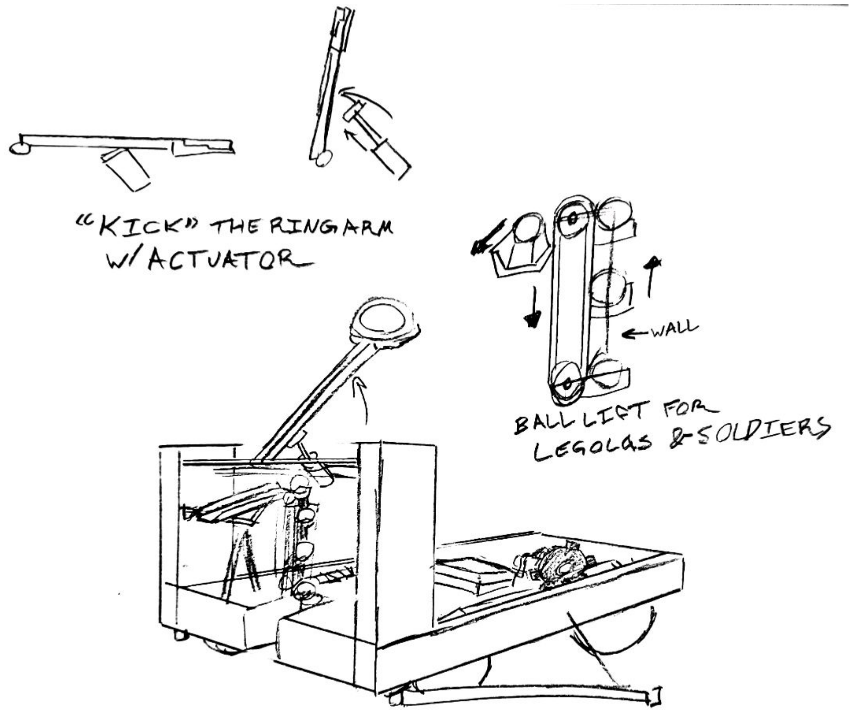


Figure 18 - Alternative Design #3, Actuator "Kick" the Ring Arm

Table 4 - Bill of Materials

	Part Description	Material	Quantity	Units	Unit Cost	Cost
1	Body	MDF	216	Square Inch	\$0.01	\$2.16
2	Wings	MDF	112	Square Inch	\$0.01	\$1.12
3	Arm	MDF	20	Square Inch	\$0.01	\$0.20
4	Arm Mounts	Plywood	60	Square Inch	\$0.05	\$0.80
5	Motor Mounts	PLA	5	Cubic Inch	\$0.82	\$4.10
6	Hinges	Steel	8		\$0.50	\$4.00
7	Screws	Steel	40		\$0.10	\$4.00
8	Axles	Steel	30	Inch	\$0.33	\$5.00
9	Axle Mounts	PLA	3	Cubic Inch	\$0.82	\$2.46
10	Legolas Launcher	MDF	25	Square Inch	\$0.01	\$0.25
11	Wheels	MDF	80	Square Inch	\$0.01	\$0.80
12	Gears	MDF	20	Square Inch	\$0.01	\$0.20
13	Soldier Tube	PVC	8	Inch	\$0.14	\$1.13
14	Body	Plywood	150	Square Inch	\$0.05	\$7.50
Total Cost						\$33.72